Atharva Bhosale

BTech
Mechanical Engineering
Indian Institute Of Technology, Tirupati

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Github LeetCode
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (ME)	Indian Institute of Technology, Tirupati	7.76	2024-2028
Senior Secondary	Vidya Valley North Point Jr College, Chinchwad HSC (MH) Board	86.33%	2024
Secondary	Podar International School, Chinchwad CBSE Board	95.8%	2022

Experience

- **Techmaniacs.** *Sept.2024 - Aug. 2026*
  Robotics Software Team Member On-Campus
  – Implemented SLAM, custom teleoperation, and hardware-interface nodes for a Rigbetel Labs TortoiseBot using Python and ROS2 (Humble).
  – Built and calibrated a drone system and developed an ArduPilot-based geofencing program for region-constrained flight.
  – Conducted robotics workshops, teaching students fundamentals of ROS2, simulation, and robot control.

Projects

- **RoboLab Web — Robotics Analysis Studio** *Oct. 2026*
  Interactive web tool for robot kinematics, dynamics, and 3D simulation Live Demo | Github
  – **Tools**: JavaScript, HTML, CSS, Three.js, Python, NumPy
  – Implemented FK/IK, DH modelling, Jacobian analysis, and 3D visualization for manipulators with up to **12 joints**
  – Added trajectory planning, RK4-based dynamics, workspace sampling up to **200,000 configurations**, and **28 automated tests**

- **Autonomous Drone Precision Landing** *Sept. 2026*
  Hybrid PID + PPO controller for autonomous landing on a moving rover in PyBullet. Github
  – **Tools**: Python, PyBullet, OpenCV, PPO, PID, Kalman Filter
  – Built ArUco-based 3D vision and Kalman-filtered rover tracking with cascaded PID control.
  – Improved landing success from **34% to 68%** and reduced mean touchdown error by **28.7%** using PPO residual learning over PID across 50 trials.

- **Omni Bot — Multi-Robot Simulation** *Sept. 2026*
  ROS 2 and Gazebo simulation for omnidirectional robot coordination. Github
  – **Tools**: ROS 2, Gazebo, Python, Ubuntu
  – Developed a configurable simulation supporting **1–2 robots** for box-face targeting.
  – Implemented configurable task parameters, 2D visualization, and optional performance graphs.

Technical Skills

- **Programming:** C/C++, Python, JavaScript, SQL
- **AI/ML & Data:** Machine Learning, Reinforcement Learning, Scikit-learn, OpenCV, NumPy, Pandas
- **Robotics & Simulation:** ROS2, Gazebo, ArduPilot, PyBullet
- **Tools & OS:** Git, Docker, Linux

Key Courses Taken

- Data Structure Algorithms, Introduction to Data Science And Machine Learning, Numerical Analysis

Positions of Responsibility

- **Club Head,** IdeaSquare tinkering and innovation hub, IIT Tirupati *Present*
- **Batch Representative,** Career Development & Guidance Council, IIT Tirupati *July. 2025 - May. 2026*
- **Core Member,** TechManiacs Tech Club, IIT Tirupati *Sept. 2024 - Aug. 2026*
- **Core Member,** SLHD Tirutsava-25 Cultural Fest, IIT Tirupati *Sept. 2024 - Feb. 2025*

Achievements

- **Secured 2nd Position** TechHeist A Capture The Flag Event, Tirutsava-25 *2025*

Certifications

- Udemy Cert on Nav2 With SLAM And Navigation, TFU|RDF|Rviz|Gazebo, Robot Operating System, Python